Thomas Gao

t25gao@uwaterloo.ca | thomasg101.github.io | github.com/Thomasg101

PROFILE

Third-year Civil Engineering student aiming at structural design across new construction, construction engineering and the renewal of existing buildings. 91.25% cumulative average, four engineering co-op terms, first-author peer-reviewed publication, and a plane-frame solver written from scratch to check commercial analysis output independently.

EDUCATION

BASc Civil Engineering (Honours, Co-operative) | University of Waterloo *Expected 2028*

Standing: 91.25% cumulative average; Term Distinction in each of the last three academic terms.

Awards: President's Scholarship of Distinction (2024) | President's Research Award (2026) | President's International Experience Award (2027).

Exchange: National University of Singapore, Fall 2026 - ranked 2nd globally in Civil & Structural Engineering, 2026 QS subject rankings.

Structural coursework: Structural Analysis CIVE 303 (88) | Solid Mechanics 1 & 2 CIVE 204/205 (93, 90) | Structure and Properties of Materials CIVE 265 (91) | Geotechnical Engineering CIVE 353 (100) | Computational Methods CIVE 121 (86).

Structural sequence ahead: CIVE 310 Introduction to Structural Design; CIVE 413 Steel Design; CIVE 414 Concrete Design; CIVE 415 Structural System Design; CIVE 422 Finite Element Analysis; CIVE 400/401 design project on a structural topic.

STRUCTURAL AND CONSTRUCTION ENGINEERING EXPERIENCE

Building Envelope Specialist (Co-op) | Entuitive

May-Aug 2026 | Vancouver, BC

- Modelled two- and three-dimensional thermal-bridge junctions across multiple project teams - slab edges, balconies, parapets, window interfaces and structural penetrations - calculating effective U-values and linear and point transmittance.
- Authored site-review reports and produced building condition assessments and 30-year capital-renewal models; identified a recurring evidence-retrieval problem and built ProjectSearch end to end, now under internal evaluation.

Engineering Technician (Co-op) | Brown Beattie Building Science Engineering

Sep-Dec 2025 | Richmond Hill, ON

- Inspected concurrent restoration sites for structural concrete repair and waterproofing, verifying surface preparation, adhesion, membrane thickness and conformance with drawings and specifications.
- Coordinated envelope rehabilitation projects from tender review through substantial performance.

Project Coordinator (Co-op) | Green Infrastructure Partners

Jan-May 2025 | Toronto, ON

- Coordinated diaphragm walls, caissons, soldier-pile-and-lagging, O-cell load testing and tieback testing for Pape Station on the Ontario Line; interpreted geotechnical and structural drawings and tracked RFIs, RFQs and submittals across the below-grade scope.
- Sequenced support-of-excavation activities and maintained field reports, quantities, testing records and as-builts through continuously changing temporary conditions.

Quality Control Technician (Co-op) | GIP Construction Engineering

2024-2025 | Thunder Bay, ON

- Ran field and laboratory quality control independently for Highway 17 and 61 rehabilitation, testing Superpave and CIREAM specimens to MTO, ASTM and AASHTO procedures.

STRUCTURAL AND COMPUTATIONAL PROJECTS

- **Plane-frame direct-stiffness solver (Python, 2026) -** Wrote a 2D frame and truss analysis tool from first principles: local element stiffness, coordinate transformation, fixed-end forces for distributed and self-weight load, moment releases, boundary-condition partition, member end-force recovery, and axial, shear, moment and deflected-shape output. Reproduces eight closed-form cases across 20 checks to machine precision.
- **Independent verification of a SAP2000 model (CIVE 303, 2026) -** Analysed a two-bay, two-storey steel sub-frame of the Mathematics 4 building under construction at Waterloo - field survey, idealisation, SAP2000 model. Re-solved it with my own code: total gravity reaction agreed to 4.7%, but the distribution to supports differed by up to 1.4 kN, and no single column section reproduced it. Traced the discrepancy to an inconsistency between the submitted property table and the model as run.
- **ProjectSearch (Entuitive, 2026) -** Local-first engineering photo and document retrieval: vision-generated records, semantic search, BM25, rank fusion, AI reranking, per-field confidence scoring and a GPS-driven 3D building view. Preserves source files and provenance; the reviewer keeps interpretation.

PUBLICATION

Gao, T., et al. (2026). *Capacity Regression and Temperature Prediction for Canada's Largest Solar Facility, Travers Solar, Alberta.* Processes, 14(7), 1078. First author. Physics-based baseline with a learned residual; R^2 0.921 on 3,077 held-out observations, leave-one-day-out validation.

TECHNICAL SKILLS

Structural analysis and design: SAP2000, S-FRAME, direct stiffness method, hand verification | **Drafting and coordination:** Revit, AutoCAD, Civil 3D, Bluebeam | **Computation:** Python (NumPy, Matplotlib, FastAPI), MATLAB, SQL, VBA | **Codes and standards:** NBC, OBC, BCBC, CSA A23.1/A23.3/S16, MTO OPSS, ASTM, AASHTO | **Building performance:** Flixo, THERM, HEAT3, WUFI